

Department of Electrical & Electrical Engineering
BRAC University
Semester- Spring 2026



EEE308L

Electronic Circuits II

Section: 06

Design Project :

Group Members:

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Submitted to

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Problem Statement:

Creating a 3rd-Order Active High-Pass Filter Design for Tweeter Protection.

Objective:

The goal is to design an active 3rd-order high-pass filter for a tweeter speaker application at Sony Corporation. The filter must effectively attenuate (reject) low-frequency signals that could potentially damage the tweeter, while allowing high-frequency audio signals above a specific threshold to pass through.

Technical Specifications:

To meet the professional standards required for speaker manufacturing, the design must adhere to the following parameters:

- **Filter Type:** 3rd-Order Active Filter (High-Pass).
- **Application:** Tweeter speaker crossover/protection circuit.
- **Cutoff Frequency f_{thres} :** Determined by the group number G using the formula:

$$f_{thres} = [10 + G] \text{ kHz}$$

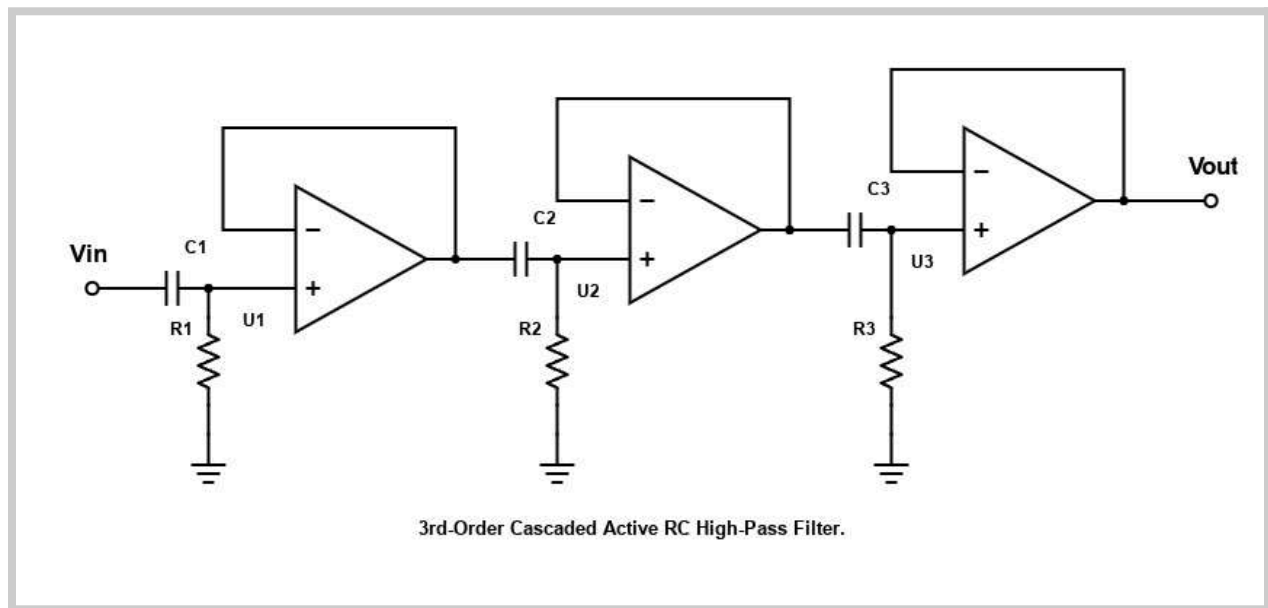
- **Performance Requirement:** The filter must reject all frequencies below the calculated f_{thres}

Design:

For our project, as our group number, $G = 4$,

Our cutoff frequency, $f_{thres} = (10 + 4)\text{kHz}$
 $= 14\text{kHz}$.

For our convenience we used a 3rd-Order Cascaded Active RC High-Pass Filter.



Reason for using this configuration instead of Sallen-key:

While the Sallen-Key topology is common in industrial designs to save space, a 3-Stage Cascaded Active Filter was selected for this project due to the following practical advantages in a laboratory environment:

1. Modular Troubleshooting:

In a 3rd-order Sallen-Key circuit, all components are part of a single complex feedback loop, making it difficult to isolate errors. By cascading three individual 1st-order stages,

the design becomes modular. This allowed for stage-by-stage verification using an oscilloscope. If the final output was incorrect, each stage could be tested independently to locate the specific point of failure.

2. Impedance Isolation (Buffering):

Passive filters often suffer from loading effects, where the second stage interferes with the performance of the first. By using an Op-Amp in each stage, we achieve high input impedance and low output impedance. This buffers the signal, ensuring that each stage operates exactly according to the theoretical formula $f = \frac{1}{2\pi RC}$ without being affected by the stages following it.

3. Predictable Stability:

High-order Sallen-Key filters rely on positive feedback, which can become unstable or "oscillate" (produce unwanted noise) when built on a breadboard due to stray capacitance. A cascaded 1st-order approach is inherently more stable and less sensitive to the wiring often found in hardware prototypes, resulting in a cleaner and more accurate frequency response.

4. Simplified Component Selection:

Achieving a specific filter shape (like Butterworth) with a Sallen-Key often requires very precise, non-standard resistor values. Cascading three identical stages allows for the use of standard lab components while still providing the required -60 dB/decade roll-off necessary to protect the tweeter.

Calculation:

Given:

- Threshold frequency = $(10 + G_v)$ kHz = 14 kHz
- Number of stages, $n = 3$

For 3-stage filter:

System cutoff frequency:

$$f_{sys} = f_c \times \sqrt{2^{\frac{1}{n}} - 1}$$

Now substitute values:

$$f_{sys} = 14 \times \sqrt{2^{1/3} - 1} = 14 \times \sqrt{1.26 - 1} = 14 \times \sqrt{0.26} = 14 \times 0.5098 \quad f_{sys} = 7.137 \text{ kHz} \approx 7137.54 \text{ Hz}$$

For each stage:

Each stage cutoff frequency:

$$f_c = f_{sys} = 7137.54 \text{ Hz}$$

Using RC formula:

$$f_c = \frac{1}{2\pi RC}$$

Rearranging for R :

$$R = \frac{1}{2\pi f_c C}$$

Given:

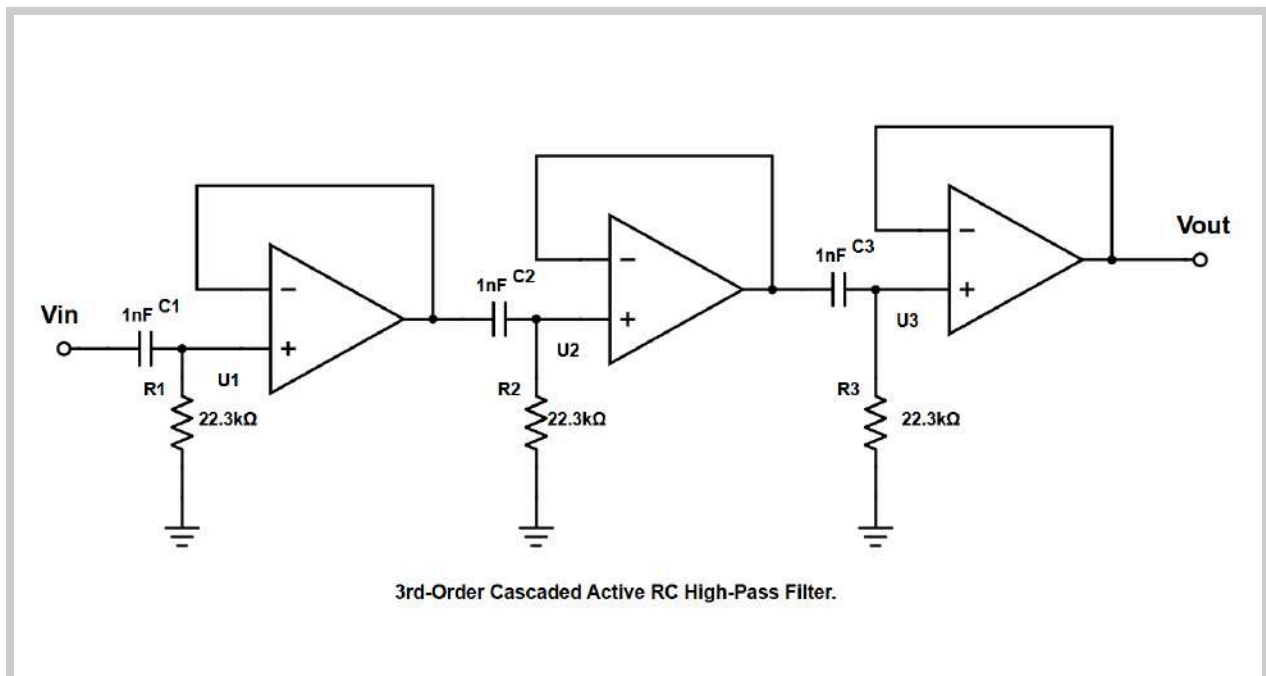
- $C = 1 \text{ nF} = 1 \times 10^{-9} \text{ F}$
- $f_c = 7137.54 \text{ Hz}$

Calculation:

$$R = \frac{1}{2\pi \times 7137.54 \times 10^{-9}} R \approx 22.3 \text{ k}\Omega$$

- Each stage cutoff frequency = **7.137 kHz**
- Component values:
 - $C = 1 \text{ nF}$
 - $R \approx 22.3 \text{ k}\Omega$

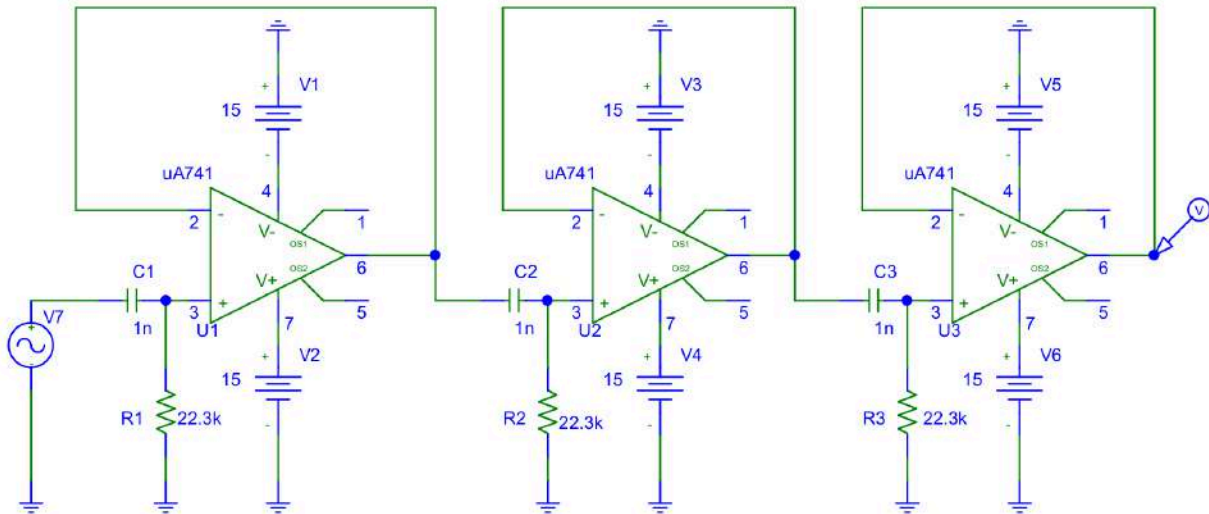
Here, as we are using an RC cascading configuration, we have adjusted the cutoff frequency for each stage using the correction formula. After our calculations we have determined the designed circuit which will look like this.



Software Implementation:

We implemented our designed circuit in PSpice schematic program

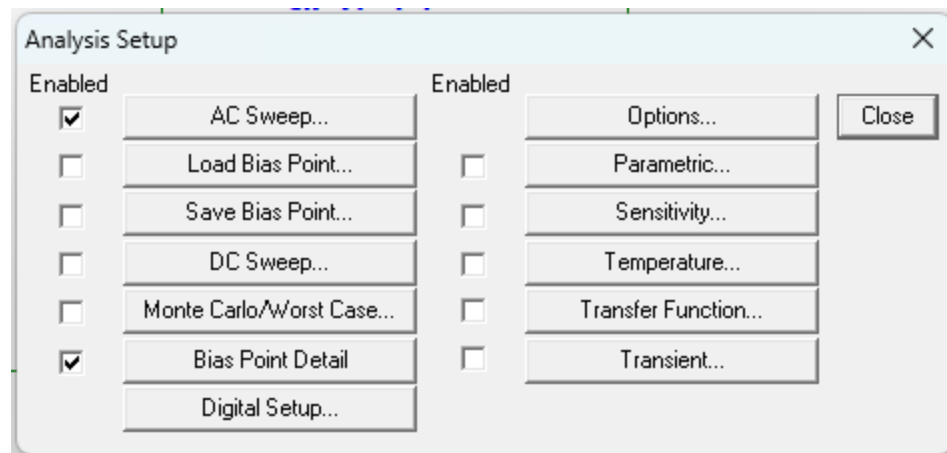
Circuit Diagram:-



Software Setup:

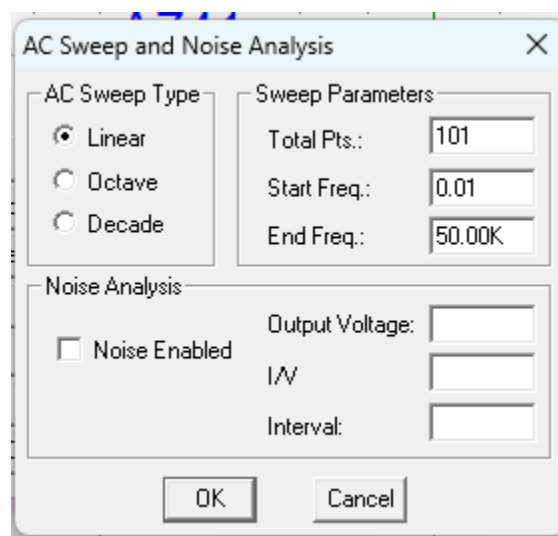
Vsin	The screenshot shows the 'V7 PartName: VSIN' dialog box. It has a 'Name' field with 'DC' and a 'Value' field with '0'. Below these is a list of attributes: DC=0, AC=1v, VOFF=0, VAMPL=1, FREQ=1k, TD=0, and DF=0. On the right side of the dialog are buttons for 'Save Attr', 'Change Display', 'Delete', 'OK', and 'Cancel'. At the bottom left are two checkboxes: 'Include Non-changeable Attributes' and 'Include System-defined Attributes', both of which are currently unchecked.
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Analysis Setup



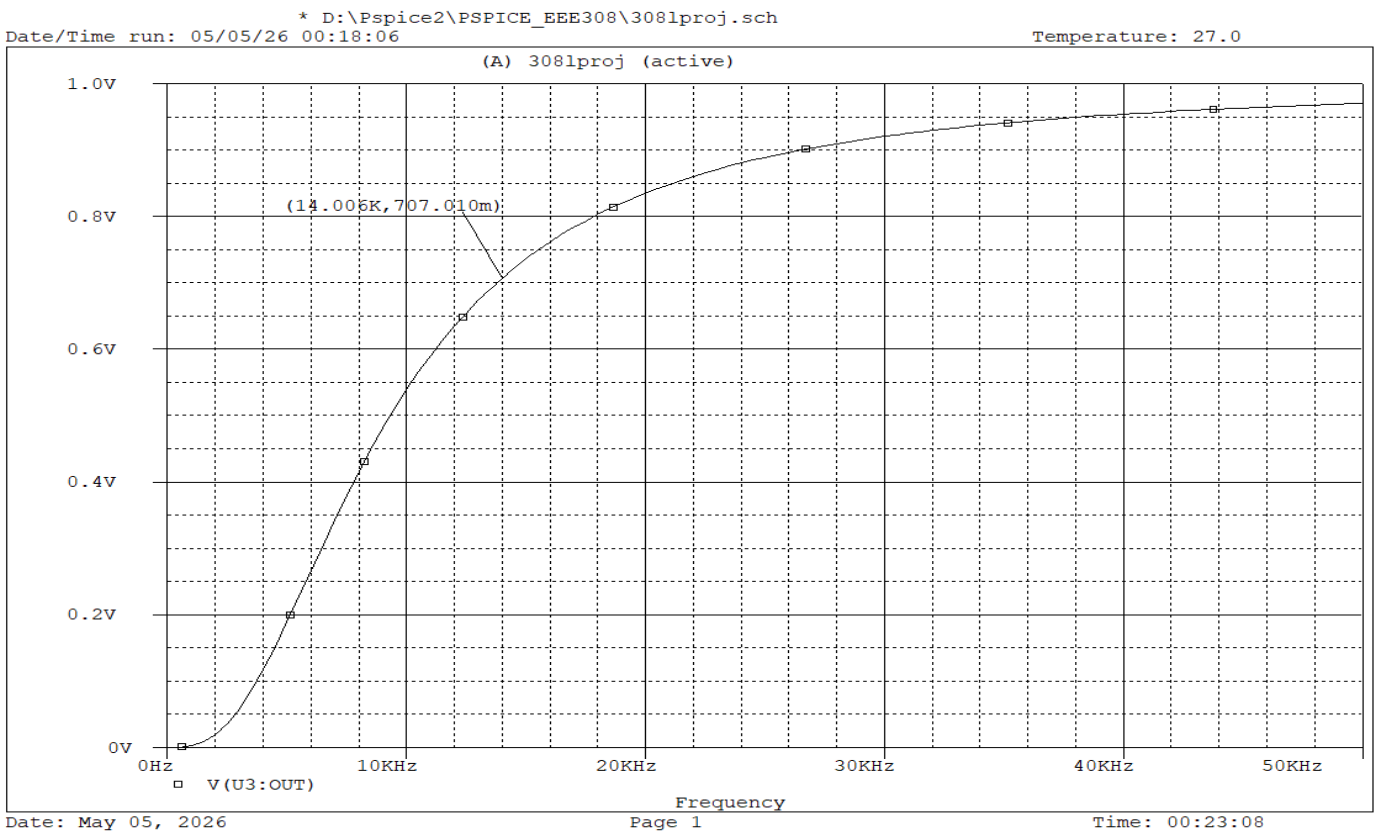
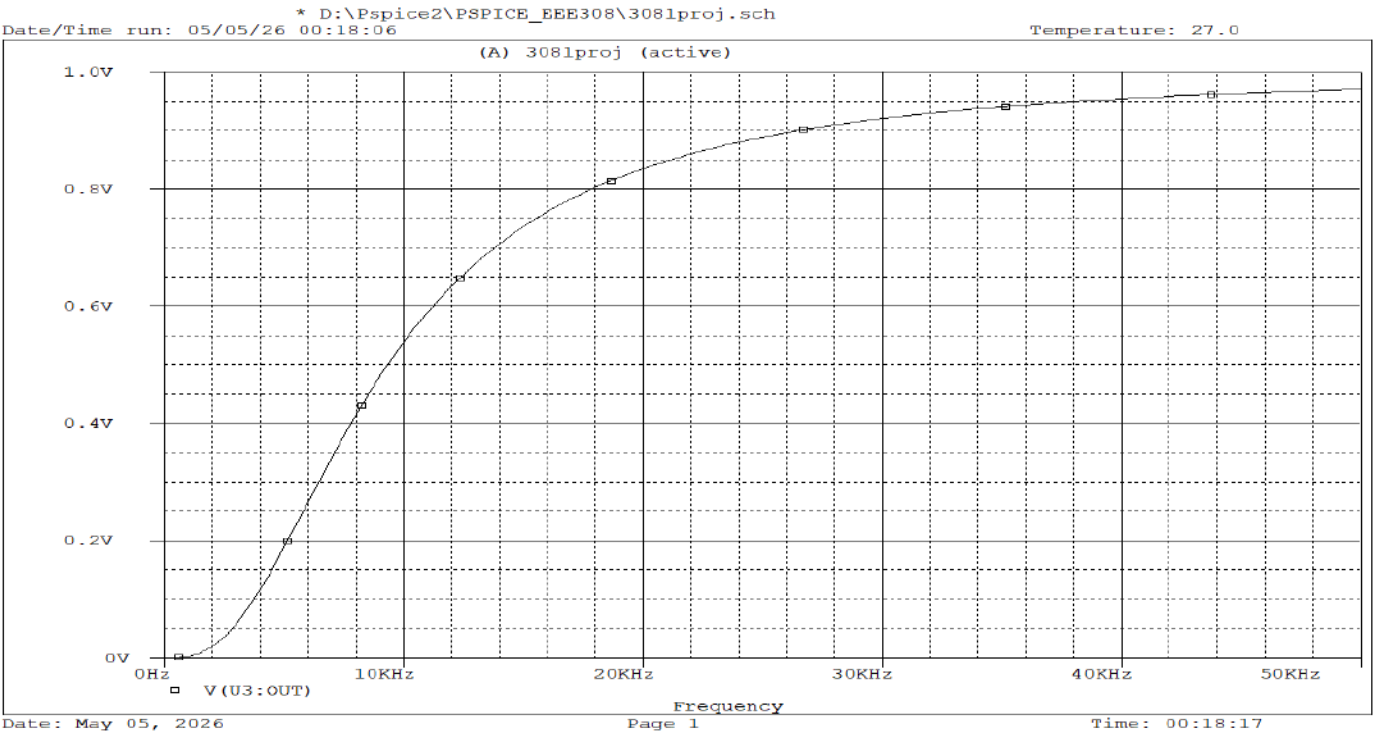
The 'Analysis Setup' dialog box is shown with a close button (X) in the top right corner. It contains two columns of buttons, each preceded by an 'Enabled' label and a checkbox. The first column has checkboxes for 'AC Sweep...' (checked), 'Load Bias Point...' (unchecked), 'Save Bias Point...' (unchecked), 'DC Sweep...' (unchecked), 'Monte Carlo/Worst Case...' (unchecked), 'Bias Point Detail' (checked), and 'Digital Setup...' (unchecked). The second column has checkboxes for 'Options...' (unchecked), 'Parametric...' (unchecked), 'Sensitivity...' (unchecked), 'Temperature...' (unchecked), 'Transfer Function...' (unchecked), and 'Transient...' (unchecked). A 'Close' button is located to the right of the second column.

AC Sweep Setup



The 'AC Sweep and Noise Analysis' dialog box is shown with a close button (X) in the top right corner. It is divided into three main sections: 'AC Sweep Type' with radio buttons for 'Linear' (selected), 'Octave', and 'Decade'; 'Sweep Parameters' with input fields for 'Total Pts.' (101), 'Start Freq.' (0.01), and 'End Freq.' (50.00K); and 'Noise Analysis' with a 'Noise Enabled' checkbox (unchecked) and input fields for 'Output Voltage:', 'I/V', and 'Interval:'. 'OK' and 'Cancel' buttons are at the bottom.

Simulation Result:



Observations:

From our simulation graphs we can see that our filter reaches a cutoff point of **0.707V** at 14.006kHz which is extremely close to our threshold cutoff frequency.

If we calculate the error,

$$\begin{aligned}\text{Error} &= \frac{14006 - 14000}{14000} \times 100\% \\ &= 0.04\%\end{aligned}$$

Hence, we can conclude saying that our software simulation was successful with an error of less than 0.1%.

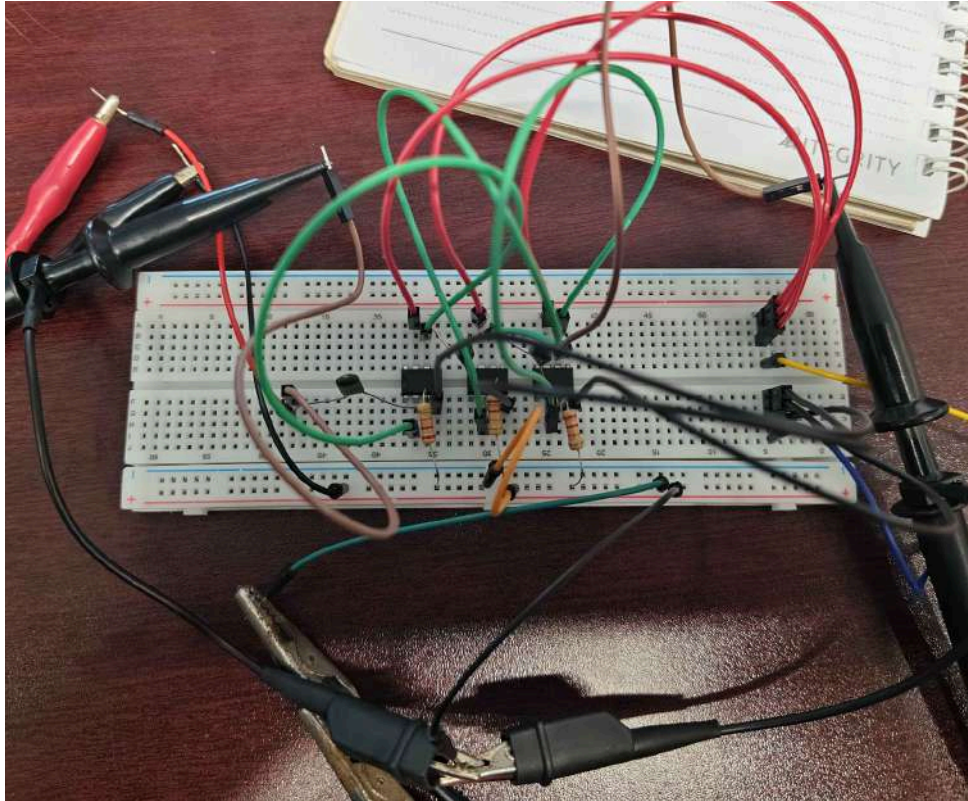
Equipments:

1. Op amp(UA741)
2. Resistor (22.3K Ω)
3. Capacitor(1nF)
4. Vsin
5. V_{DC}
6. Ground earth
7. Voltage Markers
8. PSpice Software

Hardware Implementation:

For our hardware implementation, we used the closest value available from our lab. As a resistor value of $22.3\text{k}\Omega$ isn't readily available, we used the closest value available, which is $22\text{k}\Omega$.

Circuit Diagram:-



Equipments:

1. Op amp(UA741)
2. Resistor ($22\text{k}\Omega$)
3. Capacitor($0.001\mu\text{F} = 1\text{nF}$)
4. Jumper wire
5. Breadboard
6. DC power supply
7. Oscilloscope

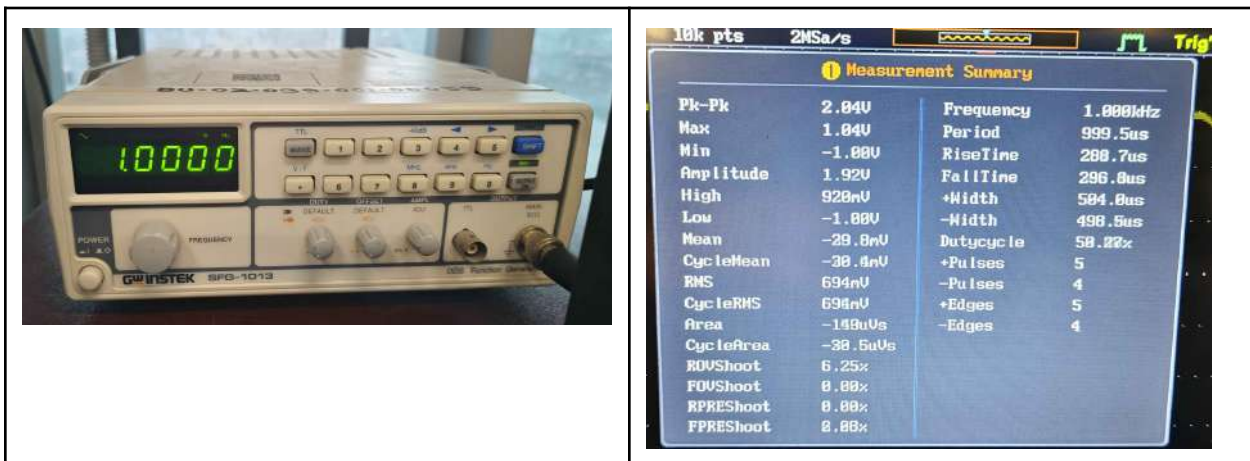
Op-amps power supply (+15V & -15V):

We used a standard DC power supply where we used the positive terminal from CH2 as our input for OP Amp terminal 7 (Positive voltage supply). For OP Amp terminal 4 (Negative Voltage Supply) we used the negative terminal from CH1.



Signal Generator:

Here, We set the input as an AC sine wave with an amplitude of 1V



We set the starting frequency at 1kHz.

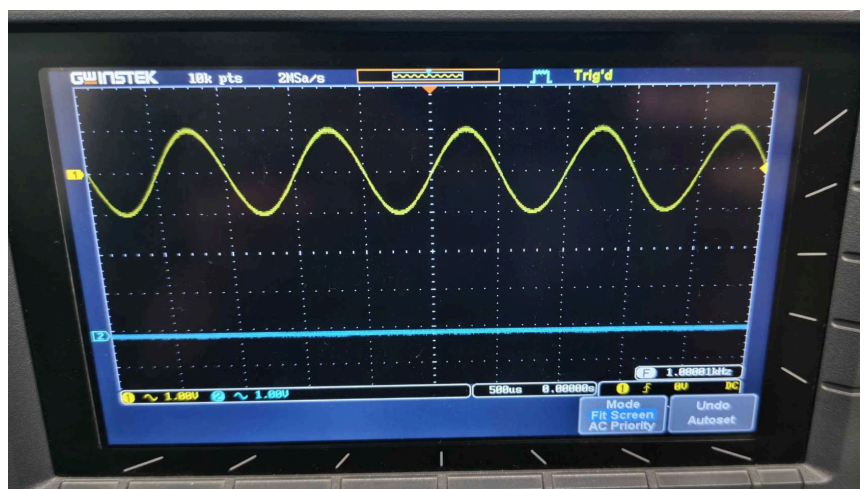
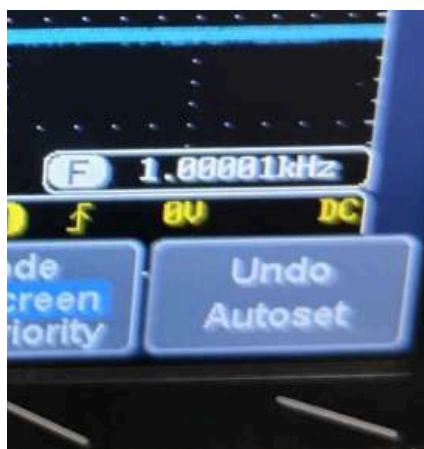
Result:

Input-Output Graph from Oscilloscope

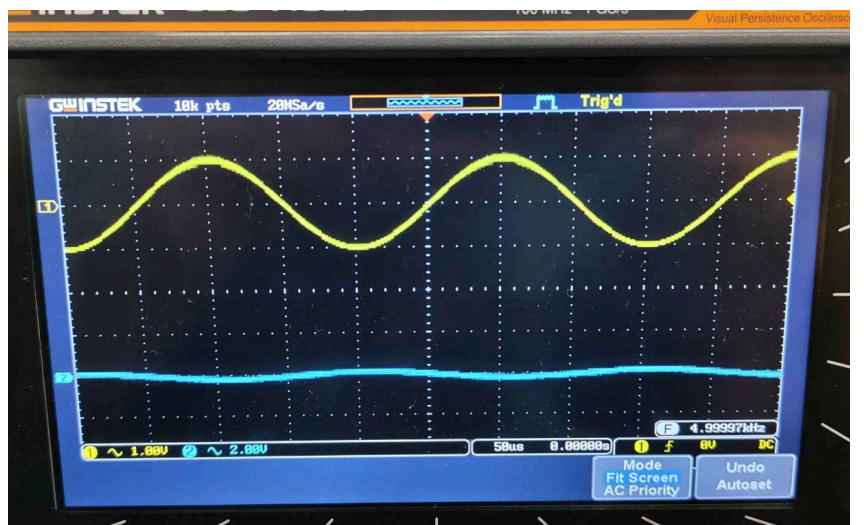
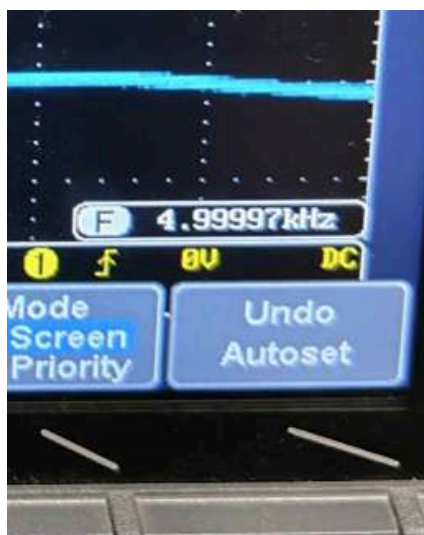
Here, Input = 1V amplitude sine wave recorded on CH1(Yellow)of oscilloscope

Output is varying which recorded on CH2(Blue) of oscilloscope

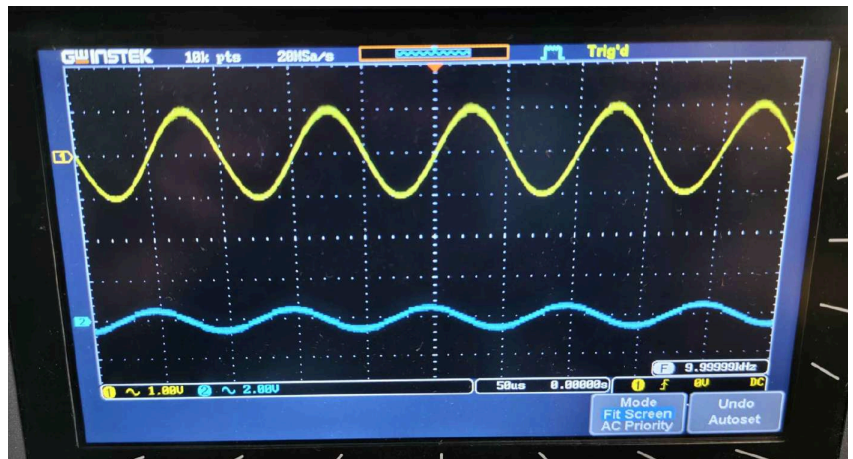
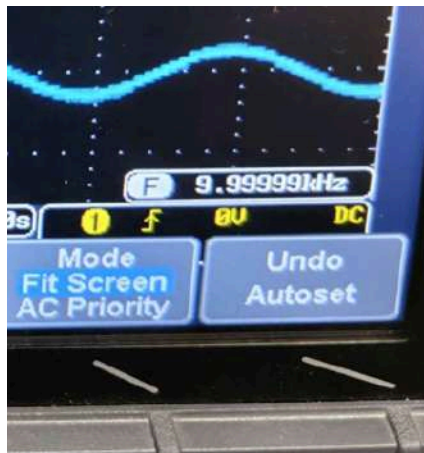
For $f = 1\text{kHz}$



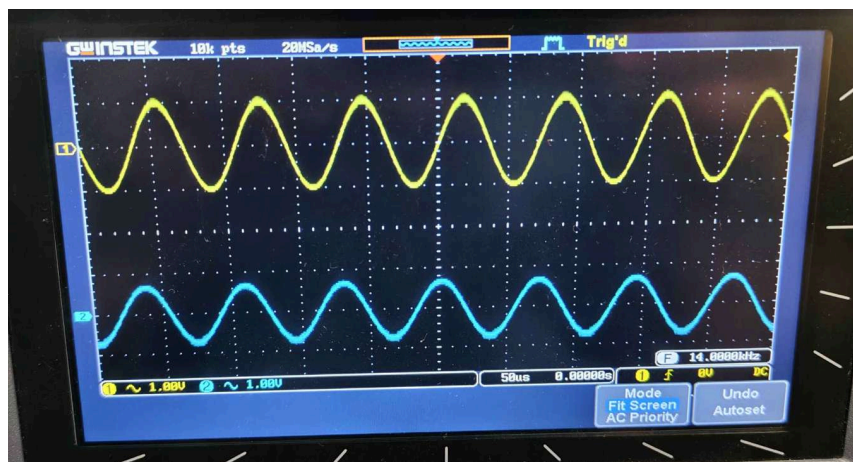
For, $f = 5\text{kHz}$



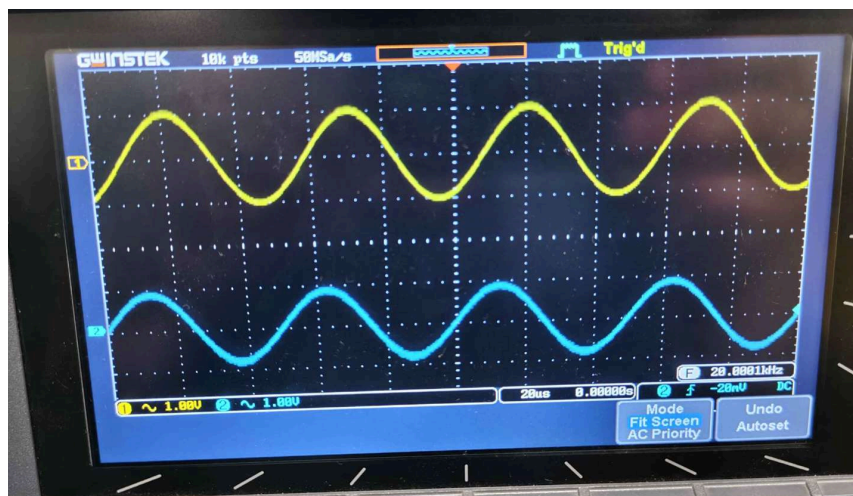
For, $f = 10\text{kHz}$



For, $f = 14\text{kHz}$



For, $f = 20\text{kHz}$



For, $f = 40\text{kHz}$

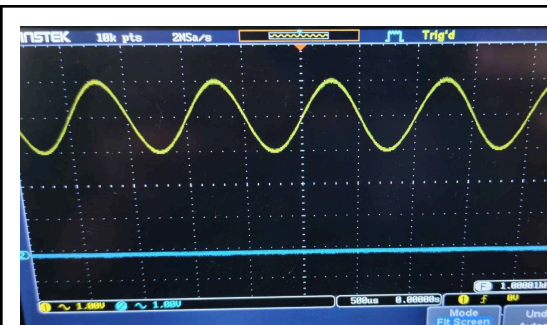


Observing the High Pass Function:

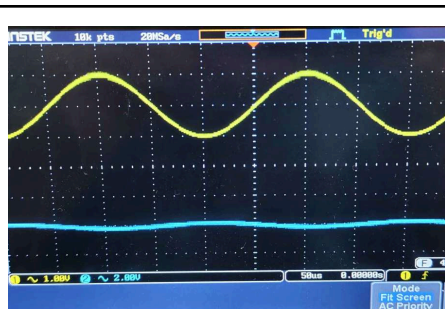
We kept eye on specific frequencies to observe the function of the filter:

At Low Frequencies:

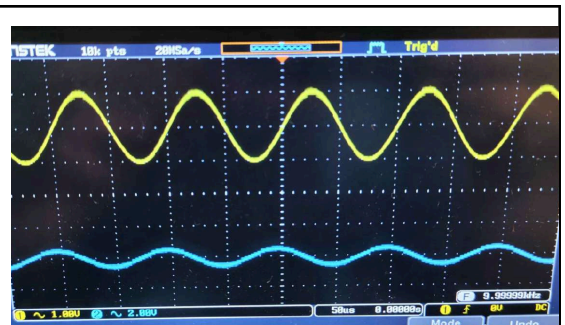
From 1kHz to 10kHz we observed that the signal from CH2 was very low, peaking at a maximum of 20-50mV. However, as the frequency was getting higher, the output started to rise as well which is evident at 1kHz. At 10kHz the output is close to 200mV.



1kHz



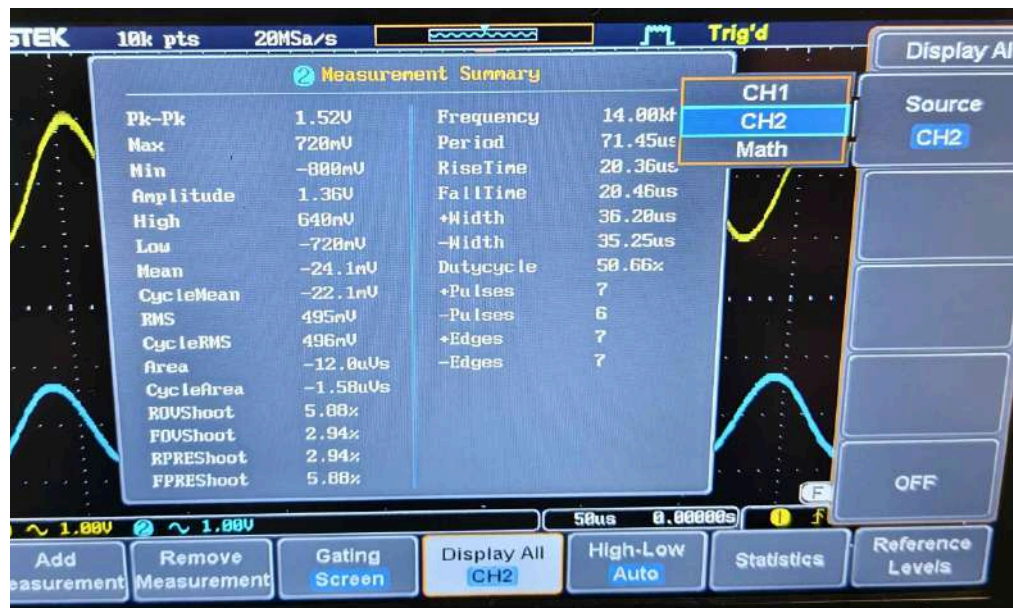
5kHz



10kHz

At Cutoff Frequency f_{thres} :

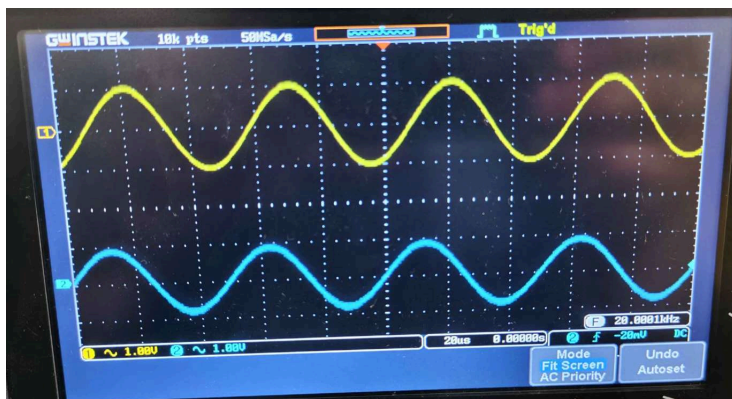
At 14kHz, we see a substantial increase in the output signal's amplitude. Using the measure function of our oscilloscope, we saw that the amplitude of voltage was close to the cutoff 0.707V.



At High Frequency:

At higher frequencies, for example 20kHz and 40kHz, we can see that the filter lets the signal pass through without any restrictions.

At $f = 20\text{kHz}$



At $f = 40\text{kHz}$



Measurement Summary			
Pk-Pk	2.84V	Frequency	40.16kHz
Max	1.00V	Period	24.90us
Min	-1.04V	RiseTime	6.952us
Amplitude	1.84V	FallTime	6.838us
High	928mV	+Width	12.34us
Low	-928mV	-Width	12.56us
Mean	-18.0mV	Dutycycle	49.56%
CycleMean	-16.5mV	+Pulses	7
RMS	677mV	-Pulses	7
CycleRMS	676mV	+Edges	7
Area	-3.61uVs	-Edges	8
CycleArea	-412nVs		
ROVShoot	4.35%		
FOVShoot	4.35%		
RPRESshoot	4.35%		
FPRESshoot	4.35%		

Error:

Hardware Error:

$$\text{Error} = \frac{720-707}{707} \times 100\%$$

$$= 1.84\%$$

Software Error:

$$\text{Error} = \frac{14006-14000}{14000} \times 100\%$$

$$= 0.04\%$$

Troubleshoot:-

During hardware implementation, several practical challenges were encountered leading to deviations between simulated and measured results. The primary issues and their resolutions are detailed below:

1. Cutoff Frequency Shift (Main Source of Error)

Issue: The simulated cutoff was 14.006 kHz, while hardware measurements showed approximately 13.5 kHz (measured at the ~ 0.707 V output point). This resulted in a 3.57% deviation from the simulation.

Cause: Use of 22 k Ω resistors instead of the calculated 22.3 k Ω due to non-availability in the lab. According to the standard cutoff formula, lower resistance inversely increases the cutoff frequency.

Solution: Acknowledged as an acceptable tolerance in real-world component selection. Future designs could use a potentiometer in series for fine-tuning, or in parallel to achieve exact theoretical values.

2. Op-Amp Malfunction

Issue: Certain stages produced no output or highly distorted signals during initial power-up.

Cause: Faulty UA741 op-amps (common in shared lab inventory) and loose dual-rail power supply connections.

Solution: Replaced damaged op-amps and verified stable dual supply (± 15 V) connections. A "stage-by-stage" testing approach was adopted before full integration.

3. Noise and Signal Instability

Issue: High-frequency noise and "jittery" waveforms were observed on the oscilloscope.

Cause: Long jumper wires acting as antennas, loose breadboard contact points, and stray capacitance.

Solution: Shortened all interconnecting wires, implemented color-coded, neat wiring.

4. Loading Effect and Amplitude Reduction

Issue: Output amplitude was slightly lower than expected across the passband.

Cause: Oscilloscope probe loading (1x setting) and the non-ideal open-loop gain characteristics of the UA741 at the upper end of the audio spectrum.

Solution: Switched to 10x probes to minimize loading and confirmed via simulation that the UA741 remains linear and effective up to 50 kHz for this specific application.

Summary of Findings:

1. Design Accuracy:

The designed 3rd-order high-pass filter achieved the target cutoff frequency of 14 kHz. Both calculation and design approach were correct and consistent.

2. Simulation Validation:

The PSpice simulation closely matched the theoretical design, giving a cutoff frequency of 14.006 kHz with a very small 0.04% error, confirming high accuracy.

3. Hardware Performance:

The practical circuit showed proper high-pass behavior.

- Low frequencies (below 10 kHz) were strongly attenuated.
- High frequencies (above 14 kHz) passed with minimal loss.

The measured cutoff output was close to the expected 0.707V, though a 1.84% error was observed due to real-world limitations.

4. Practical Challenges:

Some components, especially op-amps were damaged which caused unstable or incorrect outputs. Multiple replacements were needed to obtain correct results.

5. System Behavior:

The filter successfully protected against low-frequency signals and maintained a stable response across different frequency ranges.

6. Design Approach Effectiveness:

The cascaded structure proved to be effective, offering better stability, easy fault detection, and simple testing of each stage.

Discussion:

The primary objective of this project was the design and implementation of a 3rd-order Active High-Pass Filter to serve as a protection circuit for a high-frequency tweeter. A tweeter is highly susceptible to damage from low-frequency signals therefore, a sharp roll-off was essential. By using a 3rd-order design, we achieved an attenuation rate of -60 dB per decade, which is significantly more effective than a 1st or 2nd-order filter.

1. Topology Choice and Design Strategy

Instead of using a Sallen-Key or Butterworth active topology, we chose a three-stage cascaded RC network coupled with a uA741 Op-Amp in a voltage-follower configuration. This design was selected for several reasons:

- **Impedance Isolation:** The Op-Amp acts as a buffer, preventing the load from affecting the filter's frequency response.
- **Simplicity and Stability:** Cascading three RC stages allowed us to hit the 14 kHz target precisely while maintaining high circuit stability during the breadboard implementation.

2. Component Selection and Mathematical Calibration

The transition from theoretical calculation to hardware implementation required precise component matching:

- **Capacitance Selection:** We used a fixed value of 1 nF for all three stages. This value was chosen to keep the required resistance within a practical range for lab use.
- **The Resistance Gap:** Using the formula $R = \frac{1}{2\pi fC}$, our calculations for a 14kHz system cutoff (adjusted for 3 stages) required a resistance of 22.3kΩ. However, standard lab components only provided a 22kΩ resistor. This necessary

substitution shifted our theoretical cutoff slightly leading to a calculated hardware error of 1.84%.

3. Comparative Performance Analysis

Comparing the PSpice Simulation with the Hardware Implementation revealed high consistency:

- **Simulation Performance:** In the PSpice environment, using the exact $22.3\text{ k}\Omega$ and 1 nF values, the circuit performed almost perfectly with a cutoff at 14.006 kHz and an output of 0.707 V . This confirmed that our 3-stage cascaded model was mathematically sound.
- **Hardware Performance:** During lab testing with the $22\text{ k}\Omega$ resistors and 1 nF capacitors, at the 14 kHz threshold, the output was 0.720 V . The minimal difference between software and hardware proves the reliability of our breadboard construction.

4. Troubleshooting and Signal Integrity

Several practical issues were addressed during the lab session:

- **Low-Frequency Rejection:** We verified that at lower frequencies ($1\text{ kHz} - 10\text{ kHz}$), the output remained extremely low (peaking at $20\text{-}50\text{ mV}$). This confirmed that the filter effectively "blocks" dangerous low-end frequencies that could cause the tweeter's voice coil to overheat.
- **Power Supply Accuracy:** We ensured the Op-Amp was powered with a stable 15V DC supply. Any imbalance in the supply rails could have clipped the output signal or shifted the bias point, but proper calibration of the lab DC power supply ensured clean waveforms.
- **Signal Noise:** Initial readings showed slight noise. We resolved this by shortening the jumper wire lengths and ensuring all stages shared a common ground.

5. Final Conclusion of Design

The project successfully demonstrates that even with minor hardware component deviations (using 22 k Ω instead of 22.3 k Ω), a cascaded active filter remains the best solution for audio protection. The sharp roll-off ensures that the tweeter only receives the high-frequency spectrum it was designed to reproduce, thereby extending the longevity of the audio system.

Comparison of Results

Feature	Software PSpice	Hardware(Lab)
Cutoff Frequency	14.006kHz	14.009kHz
Output at Cutoff	0.707V	0.720V
Percentage Error	0.04%	1.84%

Video link:

[Demonstration video](#)

In case the link doesn't work

<https://www.youtube.com/watch?v=eyPUvx59Hyc>

[Software Demonstration video](#)

In case the link doesn't work

<https://youtu.be/vq0PHALZyxk>

Contribution:

Member's name	Student ID	Contribution
Mahdi Zaman	24121307	● Circuit Design. ● Calculation. ● Simulation. ● Report Preparation. ● Hardware Implementation. ● Results Observation. ● Error Analysis.
Souvik Barman Ratul	24121205	● Calculation. ● Hardware Implementation. ● Lab Apparatus Testing. ● Report Preparation. ● Trial and Error. ● Component Selection.
Afifah Bashir Sarah	23221101	● Circuit Design. ● Troubleshooting. ● Report Preparation. ● Hardware Testing.
Sadia Jahin	23221042	● Simulation. ● Circuit Design. ● Report Preparation.
Fowzia Rahman Orin	23221078	● Troubleshooting. ● Hardware Testing. ● Report Preparation. ● Logistics.